# Same item, Distinct Contexts:
# Episode-Specific Reinstatement in Free and Serial Recall

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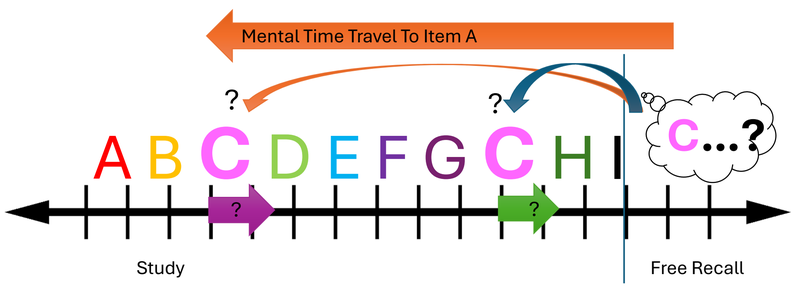
## Introduction

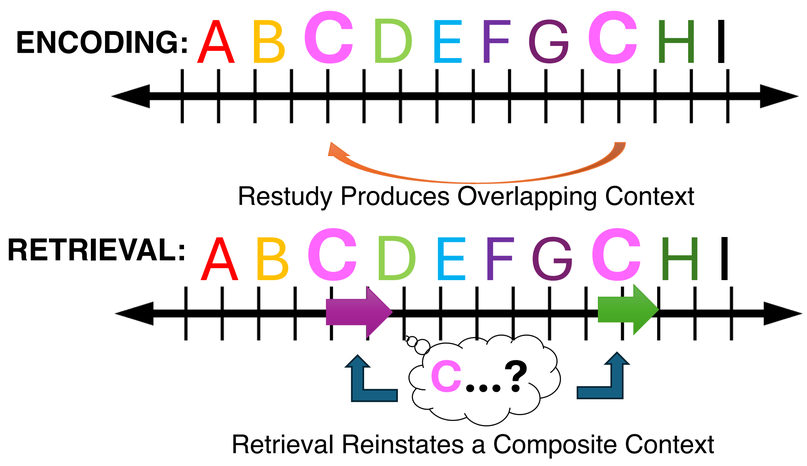
Retrieved context theory (**RCT**) explains memory search as **mental time travel**, where people revisit past contexts to retrieve associated items. Recalling an item reinstates its associated **temporal context**, biasing subsequent retrievals to items studied nearby in time. (Tulving, 1972; Howard & Kahana, 2002).




**But How Does Mental Time Travel Work for Items With Multiple Study Contexts?**

### Item-Based Reinstatement (Standard CMR)

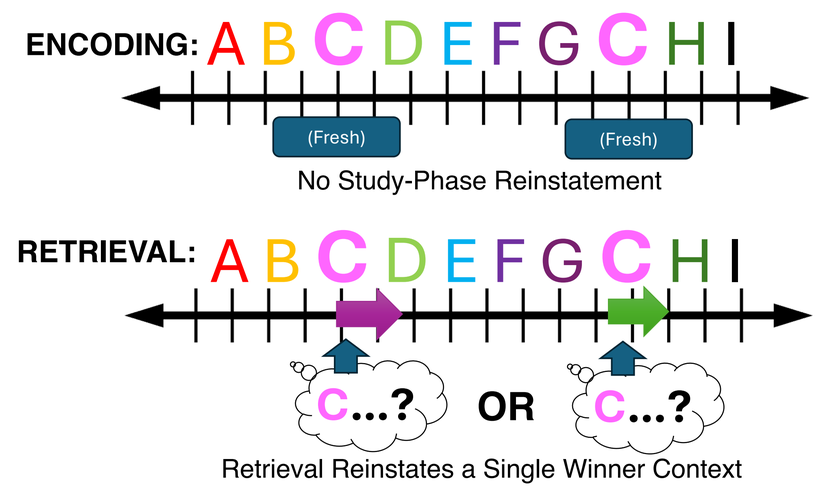
In CMR, item identity drives context evolution. Each time an item appears, it reinstates a blend of contexts associated with the item. At encoding, restudying C reinstates context that overlaps with its first presentation. At recall, retrieving C reinstates a composite of occurrence contexts that targets both presentations' neighbors for successive retrieval. Both processes **knit the two contexts together**.




**VS**

### Episode-Specific Reinstatement (Instance CMR)

In Instance-CMR, context evolves independently of items. Each occurrence encodes to a fresh, non-overlapping context. At recall, occurrence traces compete; the winner's context is selectively reinstated. Context traces are kept **distinct** and **separate**.




## Datasets

For main free recall analyses, used data from **Lohnas & Kahana (2014).**

35 participants performed delayed free recall of 48 lists presenting 40 words each, with four conditions: **control** lists with no item repetitions, **massed** repetition lists with 20 items repeated with no intervening items; **spaced** repetition lists feature 20 Items repeated with 1-8 intervening items; and **mixed** lists featuring 28 items resented once and 6 items presented twice with 0-8 intervening items.

For main serial recall analyses, used data from **Kahana & Jacobs (2000).** Twenty participants recalled 36 x 12 lists of lengths 11, 12, or 13 with two repetition conditions (no repeats vs one repeated consonant). Analyses focused on the first trial of each unique list.

Additional datasets (**Lohnas 2025**; **Kahana & Jacobs 2000**; **Broitman & Kahana, 2024**; **Howard & Kahana, 2005**) analyzed for generalization but no room here – ask if interested!

## *Free* Repetition Lag-CRP

In free recall datasets, transitions from repeaters are biased to +1 neighbor of their 1st study over 2nd vs baseline **(t= +3.009, p=.0049)**.

Standard CMR predicts the opposite pattern: more balanced transition rates to each presentation's neighbor vs baseline **(t= -2.286, p=.0286)**.

Instance-CMR predicts slim difference from baseline, while an extension that boosts first presentation traces upon repetition without knitting contexts fully captures the first presentation bias **(t=5.562, p<.0001)**.

## Identifying "True" Repetition Effects With a Position-Matched Symmetric Baseline

**Perform analyses to recall of control lists...**

**...But re-label items at same study positions as repetitions from mixed lists.**

A **good definition** of repetition effects must distinguish between **(a)** the ordinary consequences of exposing participants to additional study material and **(b)** the special consequences of encountering the same material twice instead of distinct material.

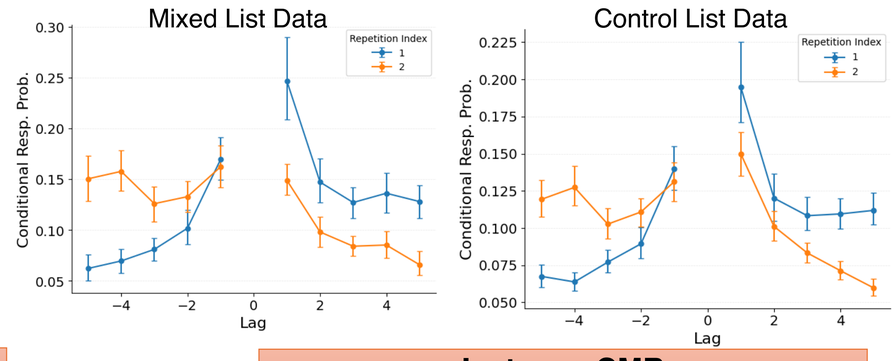
For our baseline, we focus on **(b)** and ask what would performance look like if the memory system processed the repetition in exactly the same way it would have processed a novel token, differing only in the arbitrary fact that the stimulus label is repeated? The difference between baseline and observed performance constitutes a **b**under our methodology.

For every mixed-list trial, we pair many control-list trials performed by the same participant. In each control list, we designate the items occupying the repeater's two positions as a pseudo-repeater pair. We then recompute the target statistic after relabeling those two positions as a single pseudo-item. Repeating this resampling many times yields **a null distribution** under the no-special-status hypothesis: **the memory system treats the two positions as independent study events that differ only in label.**

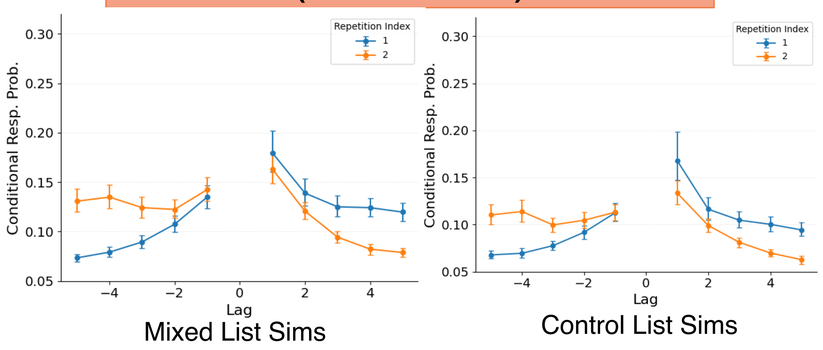
For recalls for repeated items with spacing >= 4, track **relative** to their **first** and **second** study positions:

1. For lags [-3, -2, -1, 1, 2, 3] if transition with lag is possible
2. For lags [-3, -2, -1, 1, 2, 3] if transition with lag happened

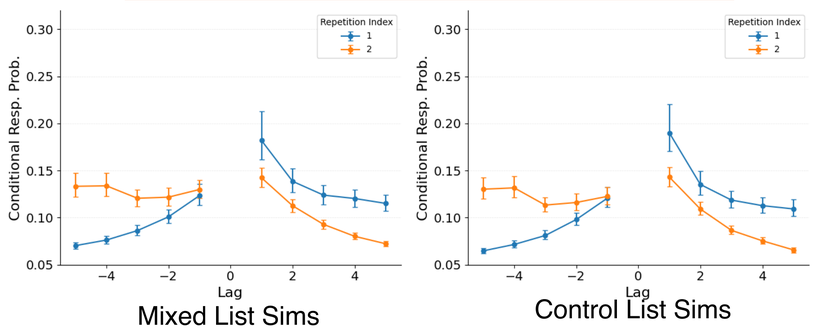
Actual Count / Possible Count = Recall Probability by Lag



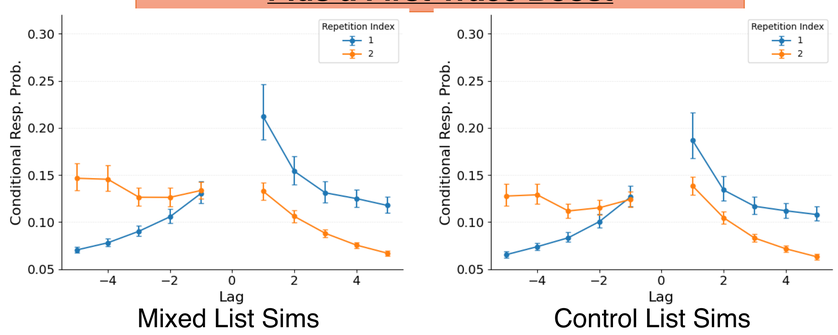

### Item-Based Reinstatement (Standard CMR)




### Episode-Specific Reinstatement (Instance CMR)




### Instance-CMR Plus a First Trace Boost



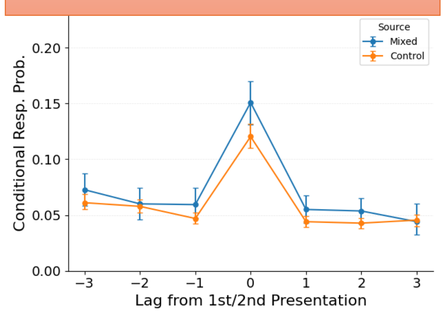

## Repetition Neighbor Contiguity

Standard CMR but not Instance CMR proposes that **neighbors** of repeated items will be associated with overlapping contexts, implying boosted transitions across neighborhoods as their associated contexts become good cues for one another.

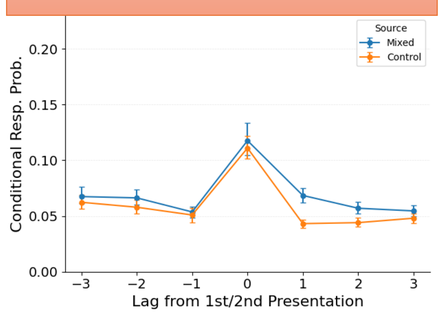
**We can adjudicate by plotting transitions from +1/+2 neighbors of the *first* presentation of repeated items to neighbors of *second* presentation, and vice versa.**

Data shows no such boosted cross-neighborhood transition rates, and instead reveals a new pattern: boosted transition rates to the repeated item itself. **Captured by Trace Boost extension.**

### Data




### Standard CMR

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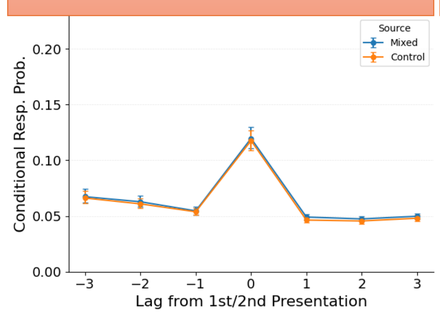


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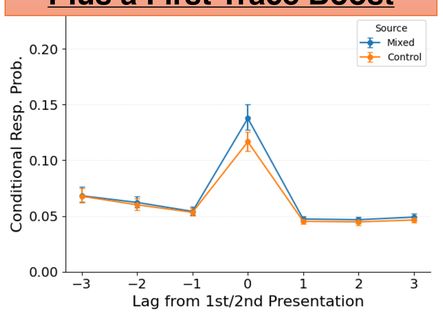
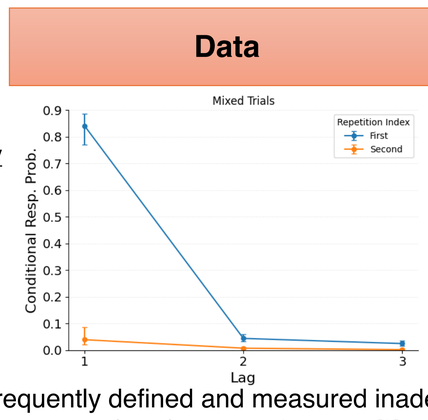



## *Serial* Repetition Lag-CRP

In serial recall datasets, there is a **correct** transition to make after recall proceeds successfully through first presentation of a repeated item: **The transition must be to the +1 neighbor of the item's 1st pres., and not to the +1 neighbor of the 2nd presentation.**

The data shows humans reliably make the correct transition and Instance-CMR successfully predicts the same absence of repetition-driven errors! But Standard CMR predicts opposite pattern: a high rate of erroneous transitions to the second presentation's neighbor.

### Data

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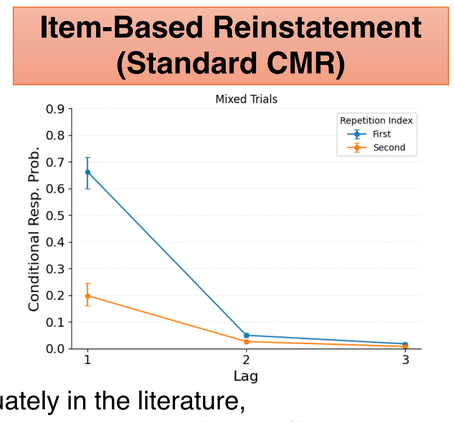


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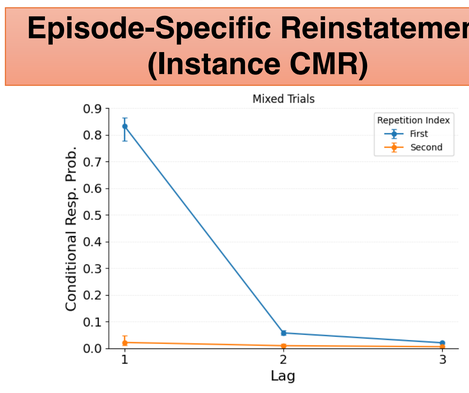


## Conclusion

| Model | Pooled ΔAIC vs Best Mean [CI] | Reliably Worse? |
|---|---|---|
| Instance CMR + Trace Boost | 0.00 [0.00, 0.00] | No |
| Instance CMR | 2.31 [-1.73, 6.35] | No |
| Standard CMR | 22.83 [15.31, 30.34] | Yes |

- Repetition effects may be frequently defined and measured inadequately in the literature, muddling interpretation. Comparison of performance to a **position-matched symmetric baseline** can distinguish real effects from natural outcome of two study positions sharing the same label.

- CMR's Item-based context reinstatement predicts patterns of **associative interference** that do not consistently arise in free or serial recall data. Occurrences may instead encode to distinct, non-overlapping contexts that compete and are reinstated separately during retrieval. With an instance-based framework, retrieved-context models can adapt to address these limitations.

- Instead of study-phase retrieval, reminding may **boost memory for an item's first presentation w/o knitting occurrence contexts**. However, model comparison outcomes so far are ambiguous.

Tulving, E. (1972). Organization of memory.
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